

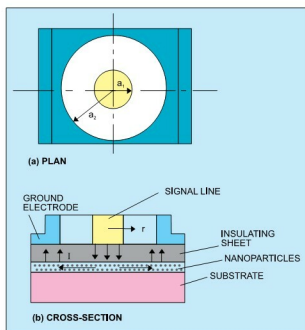


Fig. 3: Coaxial sintering head electrode

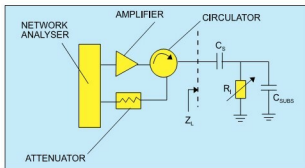


Fig. 4: Circuit of sintering setup

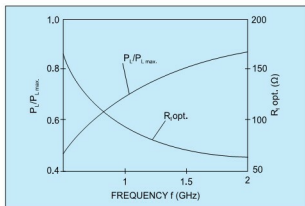


Fig. 5: Peak value of the relative power delivered to a continuous nanoparticle layer as a function of frequency 'f'

placed above the printed pattern, these capacitively couple to the printed layer, providing an approximately homogeneous electric field. A feasible coupling reactance is obtained when operating at microwave frequencies. RES of printed patterning is then carried out by moving the sintering head over the pattern or vice versa (with a fixed head over a moving substrate).

The sintering head design and optimal operating parameters depend on the layer thickness and conductivity of the nanoparticle layer, substrate properties, pattern geometry, velocity at which the pattern passes below the head and the targeted working distance of sintering head.

The initial properties of the nanoparticle layer prior to sintering depend on the materials and printing method used, as well as possible pretreatment procedures applied before sintering. Typically, with inkjet and gravure printing, the thickness of the nanoparticle layer is approximately 1 μm .

In this case the current density and heat distribution are uniform enough in vertical direction so that the geometry can be treated as 2D. When optimising for power delivery, the intrinsic reactance of the unsintered layer due to interparticle capacitance between the closely spaced nanoparticles plays some role. However, in practice, silver nanoparticle layer may be considered purely resistive throughout the process.

Fig. 2 shows a scheme of sintering electrodes above parallel lines of printed nanoparticles. The impedance seen from the sintering electrodes is the load impedance and can be represented as:

$$Z_L = R_L + jX_L$$

where R_L is the resistance of the printed line between the sintering electrodes and X_L is the reactance of coupling capacitance of sintering electrodes.

Nanoparticle layers undergo a rapid transition in conductivity during RES process causing vast variation in both R_L and X_L . Impedance matching and

power control can be optimised by measuring the power reflected from the load and by tuning the matching circuitry, frequency and input power accordingly. Fig. 2 shows 'g' as the spacing between the two electrodes and 'd' as the distance between the sintering electrodes and nanoparticle layer.

An alternative to dynamic tuning of the matching circuitry is to use several sintering heads in succession with matched impedance—i.e., the first sintering head matched to initial conductivity and so on—providing efficient power delivery throughout the process. The microwave power is distributed among the printed lines and is directly proportional to the resistance of a printed line. Hence most power is delivered to the least sintered location. This way, unlike DC RES, the AC RES setup with capacitive biasing provides uniform conductivity across printed pattern.

Coaxial sintering head

An arrangement of coaxial sintering head is shown in Fig. 3. The inner signal line has a radius (a_1) of 0.5 mm and the protective ground has a radius (a_2) of 2 mm. This arrangement of grounding prevents radiation leakage. There is an insulating sheet made of 12.7 μm thick Kapton HN polyimide film between the coaxial head and the printed pattern in order to fix working distance 'd'. When the head is brought at a working distance of 'd' above the printed nanoparticle layer, current 'I' flows through the structure and electric field 'r' is generated, as shown in Fig. 3, due to capacitive coupling of outer and inner electrodes with the nanoparticle layer.

As shown in Fig. 4, the setup also employs an amplifier feeding the sintering head through a circulator. The circulator isolates the amplifier from sintering in load impedance and directs any reflected power through an attenuator back to the network analyser for monitoring. The coaxial sintering head is motorised. Silver nanoparticle ink/nanopaste is printed on the PCB. The temperature of the PCB is kept at 60°C so that ink solvent evaporates but